

Mini Project 04

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The plot in **Error! Reference source not found.** shows the RMSE losses for 5 different predictive models for recommending movies to customers on an 80/20 train test split of existing customer data. The 5 predicted models were as follows, global mean, item mean, user mean, plain SVD, and residual SVD. The global mean method simply predicts that a user will rate a movie the same as the average rating of all movies. The item mean predicts that a user will rate a given movie the same as that movie's average rating. The user mean predicts that a

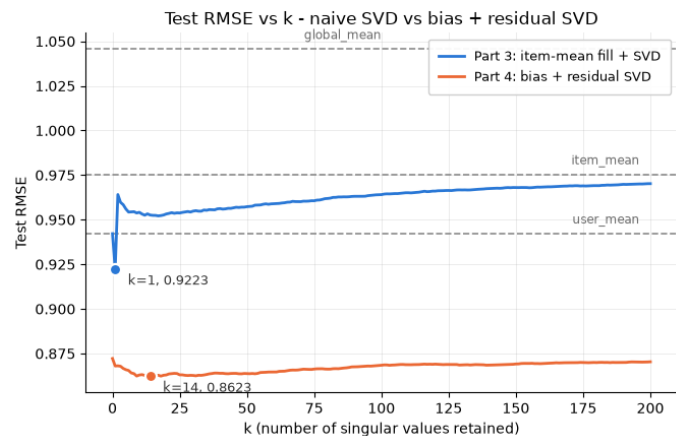


Figure 1 RMSE plot for various review prediction models

user will rate a movie the same as they've rated every other movie they've watched. The naïve SVD attempts to build a model that predicts the full score based on all user's ratings. The residual SVD model attempts to predict only the deviations in ratings from the global mean, movie mean, and user mean. This is a part of what makes the residual SVD more accurate, since the most energetic components of the naïve SVD model are spent recreating the average ratings, rather than the personalities of every user.

The most substantive factor was one separating big budget critically acclaimed films from generic crowd pleasers. People who prefer movies like The Godfather, Pulp Fiction, and Blade Runner would be drawn to films like The Empire Strikes Back, 2001: A Space Odyssey, and Fight Club. On the other side people who like The Mummy and Legally Blonde might be interested in Matrix Reloaded and Independence Day.

The limitations of this model are that it must take all the data into memory when building, making it impractical as the user base expands and more data is collected. Additionally, it assumes a linear relationship between all variables, which does not necessarily reflect how a person rates a film.

An example of a prediction made by this model is for user 414, who liked The Crow and The Secret Life of Walter Mitty, and was recommended Star Wars Episode IV. The model predicted a rating of 4.25 and they scored the film a 5 (held out from training set)

With an additional two weeks, I would take time to train more advanced models on this data that do not require storing the whole dataset in memory the entire time. These models would allow for a comparison to be made and potentially better predictions. A more advanced model might allow for it to capture a more nuanced relationship between the movies a person watches and how they might rate a new film.